

SleepSpace AI: Revised Project Dashboard

Optimizing Machine Learning workflows from Google Colab to a Professional Multi-page Streamlit Dashboard with research-backed insights.



Kelompok Tugas Besar Data Science

Responding to **Asprak's Feedback**

A strategic pivot towards professional standards, clear scale explanations, and dashboard modularity.

Research & Data Deskwork

Deskwork Source

Analysis of 15,000 synthetic records based on clinical lifestyle research to ensure realistic sleep-stress correlation patterns.

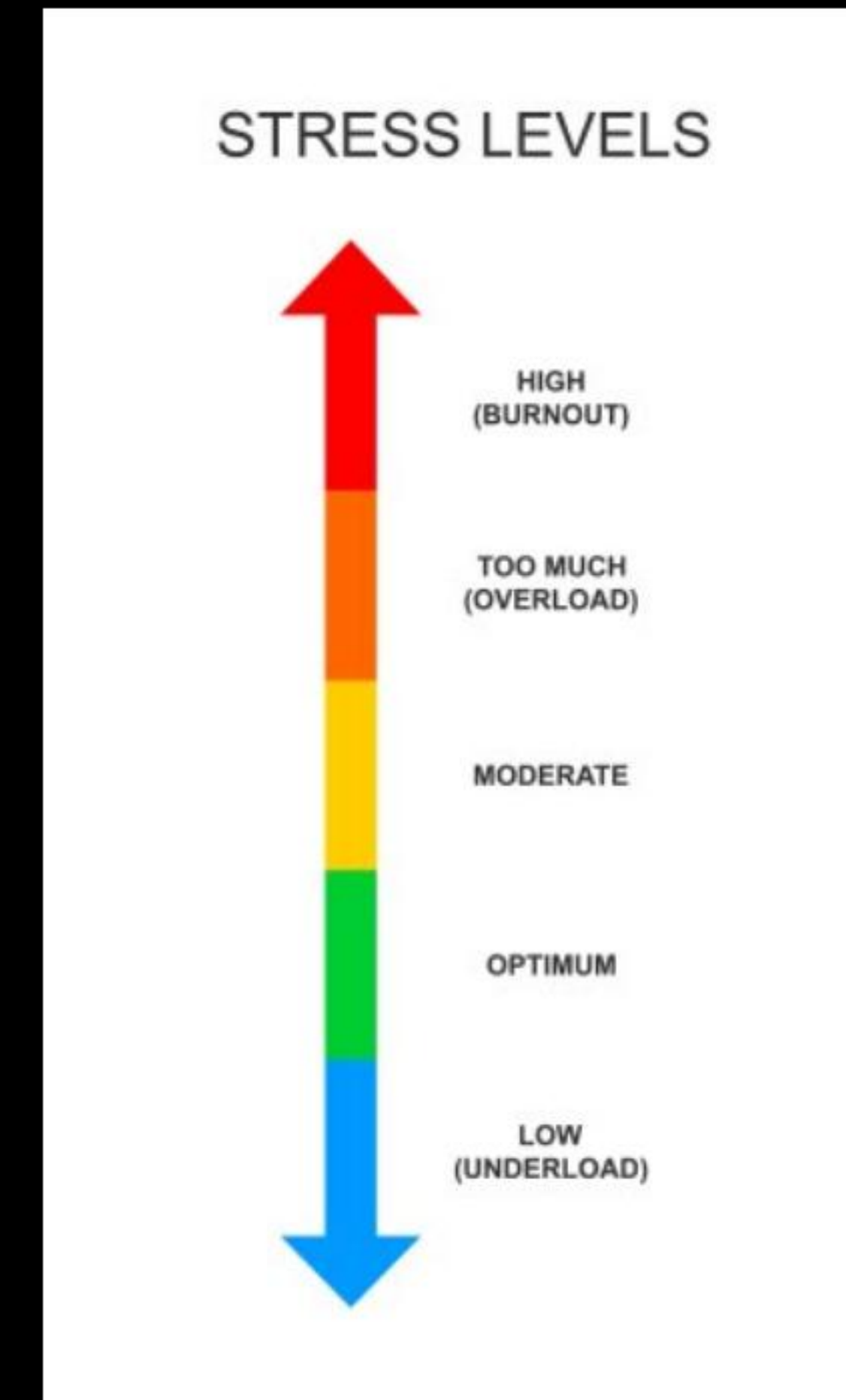
Core Variables

Focusing on Screen Time, Stress Levels, and Mental Fatigue as the three primary pillars of sleep quality disruption.

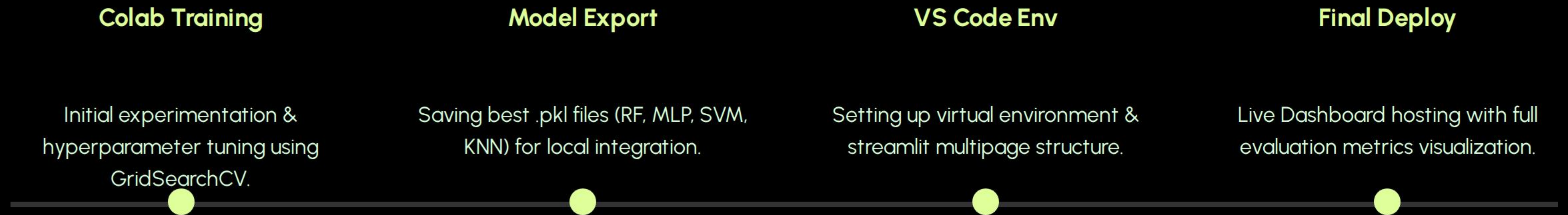
Brain Wave Frequencies		
Type & Range		What it Does
Gamma Waves HIGHER THAN 30 Hz		While concentrating, focusing and learning
Beta Waves 30 - 13 Hz		During most activities while awake
Alpha Waves 8 - 12.99 Hz		While relaxed or sleepy
Theta Waves 4 - 7.99 Hz		During stage 2 & 1 (light) sleep
Delta Waves 1 - 3.99 Hz		During stage 3 (deep) sleep

Feature Scales Glossary

-  **Stress Level (1-10)**
1-3: Low (Minimal effect), 4-7: Moderate (Common daily stress), 8-10: Severe (High sleep impact).
-  **Mental Fatigue (1-10)**
1-3: Alert, 4-7: Tired (Reduced focus), 8-10: Exhausted (Clinical burnout symptoms).



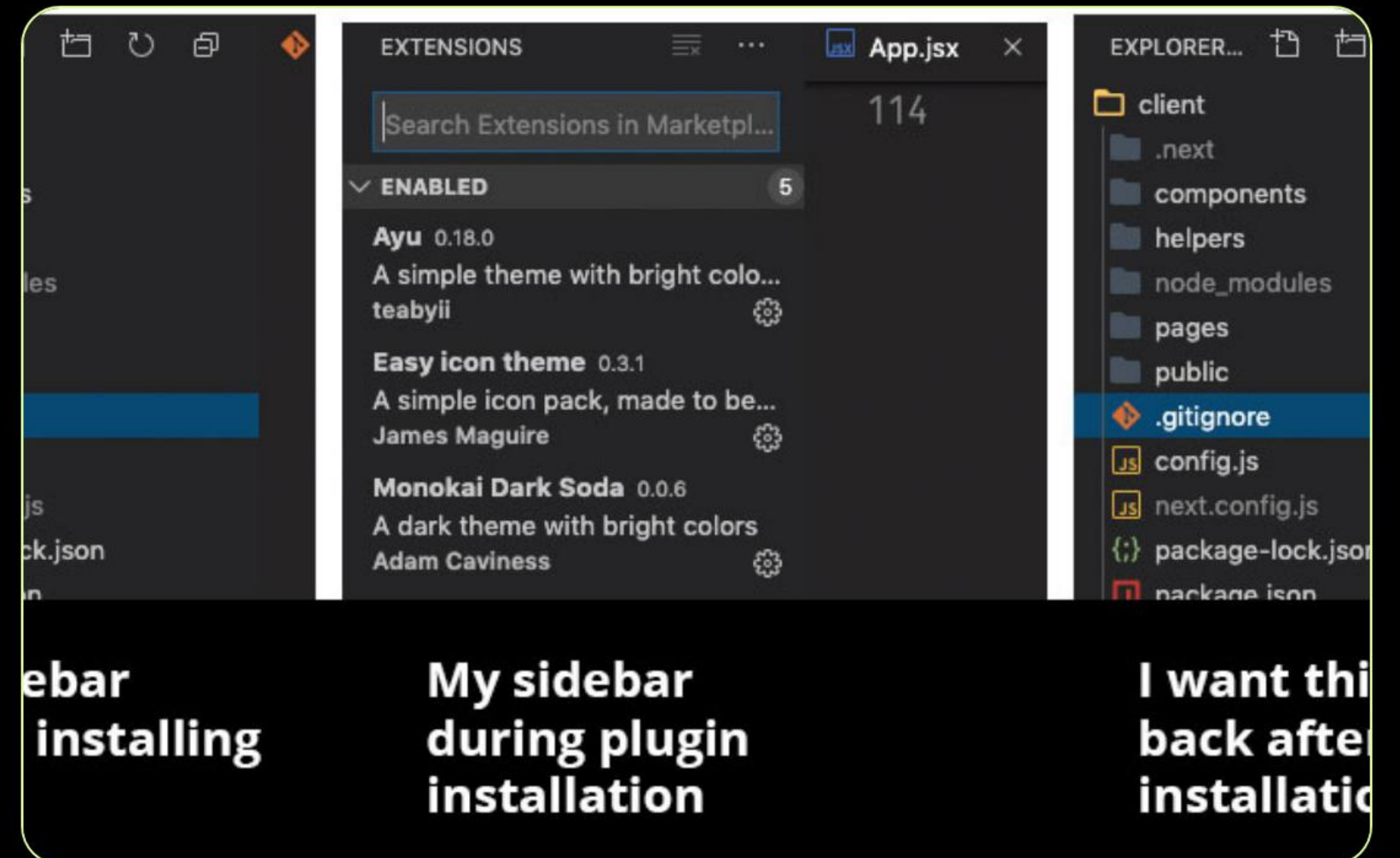
Workflow: Colab to Web



Multi-page Sidebar Structure

Our `app.py` acts as the orchestrator for the sidebar navigation system:

-  **/pages Directory**
Automatically populates the sidebar with navigation links.
-  **1_Dashboard.py**
Overview & main metrics.
-  **2_Evaluation.py**
Model comparison tables.
-  **3_Prediction.py**
User-facing input panel.



The **Multi-Model** Lab



Anisa

Random Forest (Best
Performance)



Ataya

Neural Network (MLP
Classifier)



Aqila

SVM (High Recall Focus)



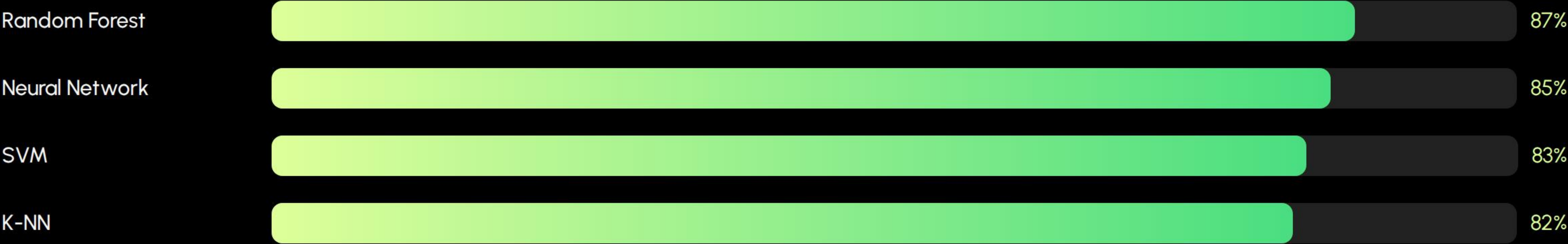
Zahara

K-NN (Baseline Logic)

Evaluation Benchmarking

Model Name	Accuracy	Precision	Recall	F1-Score
Random Forest (Anisa)	87%	42%	74%	0.54
Neural Network (Ataya)	85%	41%	70%	0.53
SVM (Aqila)	83%	38%	93%	0.52
K-NN (Zahara)	82%	35%	65%	0.48

Performance Summary

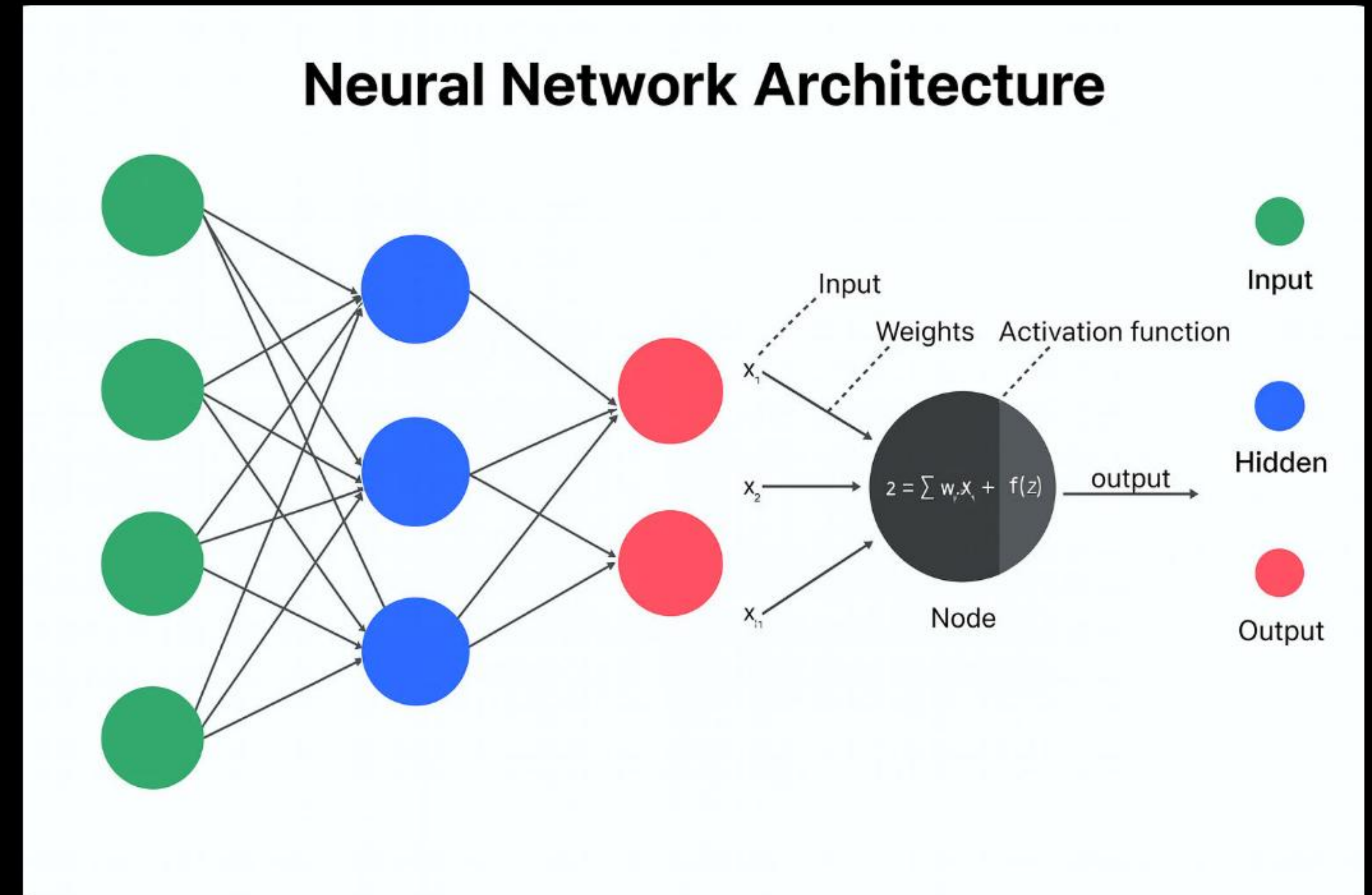


Random Forest consistently achieves the best balance between precision and recall across all testing data.

MLP Complexity Analysis

Neural Network Training

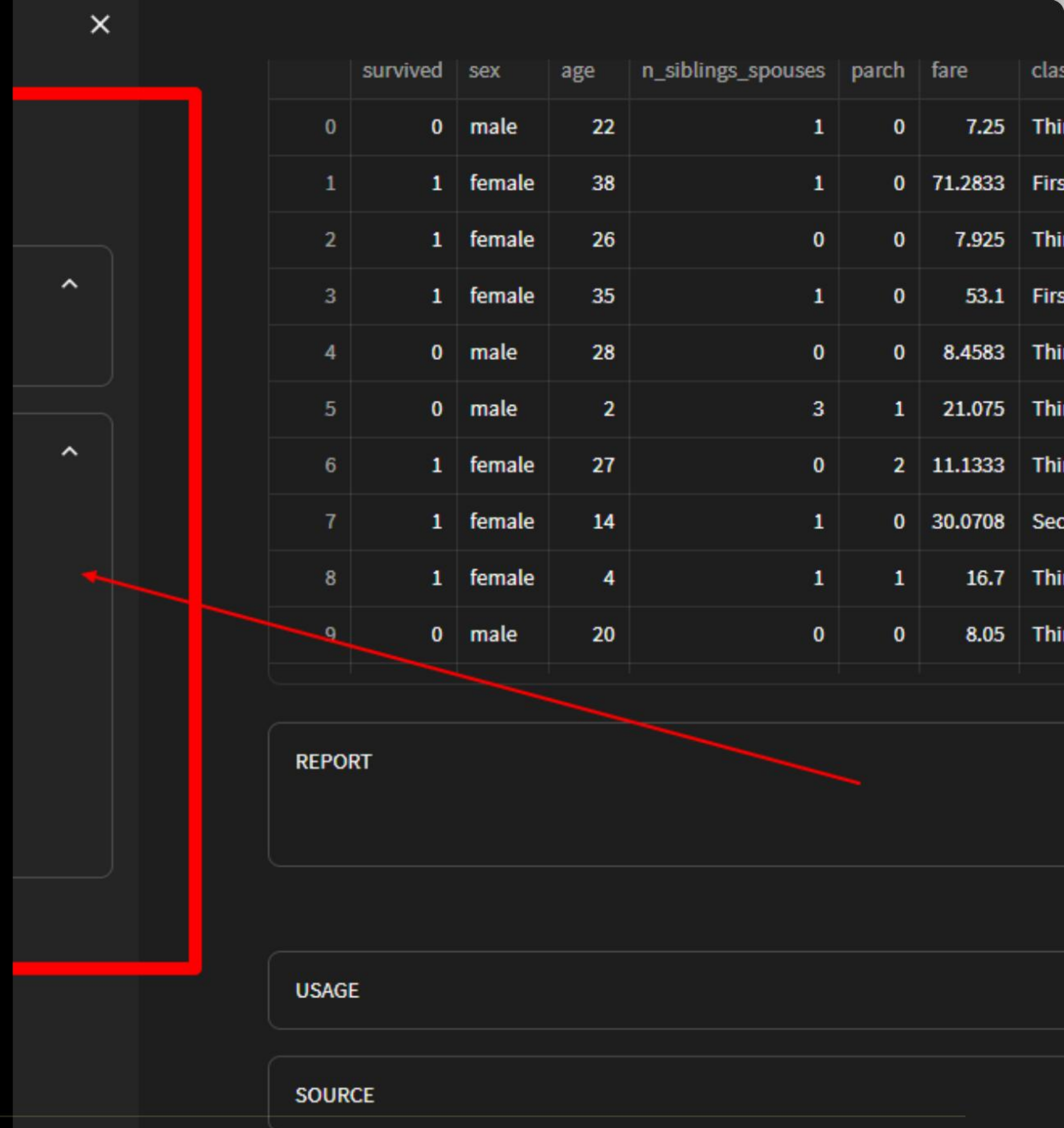
Ataya's MLP implementation uses ReLU activation and Adam optimizer. The chart highlights the convergence of training loss over 100 epochs during the Google Colab phase.



Sidebar Navigation UI

Our dashboard implements a clean sidebar for seamless user experience.

-  **Landing Page**
Context & Research overview.
-  **Model Insights**
Training results & confusion matrices.
-  **Prediction Center**
Real-time sleep quality analysis.



Questions ?

Thank you for your guidance, Asprak!

Kelompok Tubes Data Science

Zahara | Anisa | Ataya | Aqila



Image Sources



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